

Type: VSC2 (Vocational and Skill Enhancement Courses) B.Sc(ECS)-II (Semester III) Course Title: Based on DSC2-3 and DSC2-4 (Paper Code:)	
Total Lectures: 60 Hrs. Contact Hrs. (P) : 4	
University Evaluation: 30 Marks Internal Evaluation: 20 Marks	
1.	Write a Python program to calculate mean, median, mode, variance, standard deviation, range, and interquartile range (IQR).
2.	Visualize data using histograms and scatter plots.
3.	Solve numerical problems involving measures of dispersion using Python. Compute and compare ranges, variances, standard deviations, and coefficients of variation for given datasets. Interpret results to conclude data variability and distribution.
4.	Develop Python functions to compute measures of dispersion: I. Range for ungrouped and grouped data II. Variance and Standard Deviation (SD) for ungrouped and grouped data III. Coefficient of Variation (CV) to compare variability across different datasets
5.	Implement Python functions to calculate and classify types of correlation: positive, negative, and zero correlation.
6.	Write a Python script to plot a scatter diagram when $r = -0.85$, $r = 0$ and $r = 0.85$.
7.	Write Python scripts to verify basic assumptions underlying the correlation coefficient (linearity, nonlinearity, etc.). Solve numerical problems involving correlation coefficient calculations and interpretation.
8.	Implement Python functions to calculate regression line of x on y using the method of least squares.
9.	Implement Python functions to calculate the regression line of y on x using the method of least squares.
10.	Build a simple linear regression model to predict outcomes (e.g., predict student scores based on study hours).
Practical on Advanced Python Programming	
1.	Write a program to demonstrate higher order functions.
2.	Write a program to demonstrate type hinting.
3.	Write a program to demonstrate recursive functions.
4.	Write a program to demonstrate iterators and generators.
5.	Write a program to demonstrate the use of *args and **kwargs arguments.
6.	Write a program to demonstrate difference between shallow copy and Deep copy.
7.	Write a program to demonstrate various functions of RegEx.
8.	Write a program to demonstrate functions of match object.
9.	Write a program to demonstrate parsing dict to JSON string.
10.	Write a program to demonstrate parsing dict to JSON file.
11.	Write a program to demonstrate parsing JSON string to dict.
12.	Write a program to demonstrate parsing JSON file to dict.

13.	Write a program to read CSV file to dict.
14.	Write a program to write CSV file from dict.
15.	Write a program to create simple spreadsheet and do the following operations- a) Adding cell values b) Updating cell values c) Insert row d) Delete row e) Insert column f) Delete column
16.	Write a program to create simple spreadsheet and do the following operations to manage the sheet- a) Freezing rows and columns b) Adding filter c) Adding formulaes d) Adding styles e) Adding images f) Adding barchart